

Learning with Unknown Input Observers for Robust Nonlinear Estimation

A hybrid, certified estimator for systems with unknown dynamics

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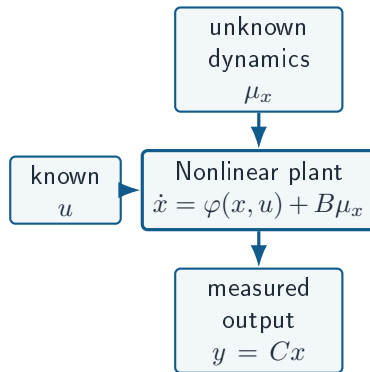
Core idea: use a bounded unknown-input observer as a reliable teacher for a neural adaptive observer.

Why estimation becomes difficult with unknown dynamics

Nonlinear plant with an unknown input

$$\begin{aligned}\dot{x} &= \varphi(x, u) + B\mu_x, & x &\in \mathbb{R}^{n_x}, \\ y &= Cx, & \mu_x &\in \mathbb{R}^{n_\mu}.\end{aligned}$$

- ▶ $\varphi(x, u)$: known nonlinear physics;
- ▶ μ_x : unmodelled dynamics, disturbances, or force residuals;
- ▶ only y is measured and μ_x must be inferred online.



Goal: estimate x and infer μ_x .

The classical UIO condition is often too restrictive

Direct unknown-input decoupling requires

$$\text{rank}(CB) = \text{rank}(B).$$

- ▶ This guarantees that the output is informative enough to isolate μ_x .
- ▶ In real vehicle and robotic systems, CB can be rank deficient.
- ▶ Enforcing exact decoupling can then rule out an otherwise useful observer.

$B\mu_x$ changes the state

$CB\mu_x$ may not
be visible in y

Relax exact decoupling;
certify a finite error bound

Regularization converts a rank obstacle into a bounded disturbance

$$\begin{aligned}\dot{x} + E_{\delta_1} \dot{\mu}_x &= \varphi(x, u) + B\mu_x + E\omega_{\delta_1}, \\ y &= Cx + D_{\delta_2} \mu_x + D\omega_{\delta_2}, \\ E_{\delta_1} &= \delta_1 E, \quad D_{\delta_2} = \delta_2 D, \\ \omega_{\delta_1} &= \delta_1 \dot{\mu}_x, \quad \omega_{\delta_2} = -\delta_2 \mu_x.\end{aligned}$$

- ▶ The transformed model is equivalent to the original plant.
- ▶ The tunable δ_1, δ_2 restore observer design freedom.
- ▶ The price is explicit: ω_δ enters the performance analysis.

Certified consequence

$$\|\tilde{\zeta}\|_{\mathcal{H}^1} \leq \lambda_\delta \|\mu_x\|_{\mathcal{H}^1} + \beta \|\xi_0\|.$$

Exact convergence is recovered when the classical rank condition holds.

Output derivatives create a generalized unknown-input observer

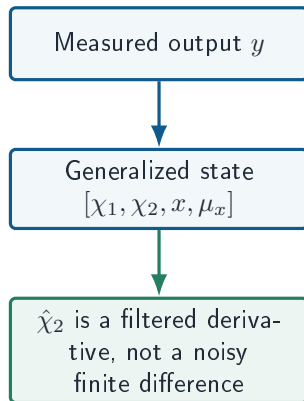
Introduce $\chi_1 = y$ and $\chi_2 = \dot{y}$ in the augmented state $\zeta = [\chi_1^\top \ \chi_2^\top \ x^\top \ \mu_x^\top]^\top$:

$$\dot{\chi}_1 = \chi_2,$$

$$\dot{\chi}_2 = C\bar{\varphi}_x(x, u) + C\frac{\partial\varphi}{\partial x}(x, u)B\mu_x + CB\dot{\mu}_x,$$

$$\dot{x} = \varphi(x, u) + B\mu_x,$$

$$y_\chi = \chi_1 + C_\chi\chi_2 + Cx.$$



Layer 1: LMI-certified generalized UIO

Observer realization

$$\begin{aligned}\dot{\eta} &= P_{\zeta} \varphi_{\zeta}(\hat{\zeta}, u) + K(y - \mathcal{H}\hat{\zeta}), \\ \hat{\zeta} &= \eta + Q_{\zeta} y, \quad P_{\zeta} \mathcal{E} + Q_{\zeta} \mathcal{H} = I.\end{aligned}$$

$$K = \mathcal{P}^{-1} Q^{\top}.$$

Guarantee

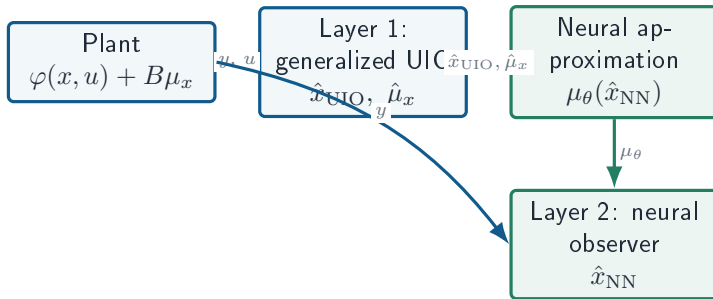
Feasibility yields an $\mathcal{H}^1$ estimation bound for the state and unknown input.

For every vertex j of the nonlinear embedding, solve

$$\begin{bmatrix} \Xi_j & \Gamma_j & \Theta_j \\ \Gamma_j^{\top} & -\lambda I_{2n_{\mu}} & \Gamma_j^{\top} \\ \Theta_j^{\top} & \Gamma_j & -\epsilon^{-1} \mathcal{P} \end{bmatrix} \prec 0.$$

- ▶ offline convex synthesis;
- ▶ online observer with a quantified robustness level λ_{δ} .

A certified teacher supervises learning



Teacher: bounded supervision

Student: state-dependent residual learning

Layer 2: neural adaptive observer with an $\mathcal{L}_2$ guarantee

Neural observer

$$\begin{aligned}\mu_\theta(\hat{x}_{\text{NN}}) &= \mathcal{N}_\theta(\hat{x}_{\text{NN}}, \hat{x}_{\text{UIO}}, \hat{\mu}_x, u), \\ \dot{\hat{x}}_{\text{NN}} &= \varphi(\hat{x}_{\text{NN}}, u) + B\mu_\theta(\hat{x}_{\text{NN}}) + L_{\text{NN}}(y - C\hat{x}_{\text{NN}}).\end{aligned}$$

$$\begin{aligned}\mathcal{L} = & \|\hat{x}_{\text{NN}} - x_{\text{ref}}\|_{T_{\text{ref}}}^2 \\ & + \|y - C\hat{x}_{\text{NN}}\|_{T_y}^2 \\ & + \|\hat{x}_{\text{NN}} - \hat{x}_{\text{UIO}}\|_{T_{\text{UIO}}}^2 \\ & + \lambda \|\mu_\theta - \hat{\mu}_x\|^2.\end{aligned}$$

With $\Delta_x = \mu_x - \mu_\theta(\hat{x}_{\text{NN}})$,

$$\|\epsilon_{\text{NN}}\|_{\mathcal{L}_2} \leq \lambda_{\text{NN}} \|\Delta_x\|_{\mathcal{L}_2} + \beta_{\text{NN}} \|\epsilon_{\text{NN}}(0)\|.$$

$$\begin{bmatrix} (A_j^x)^\top P + PA_j^x & PB \\ -C^\top Q - Q^\top C + I & \\ B^\top P & -\sigma I \end{bmatrix} \prec 0$$

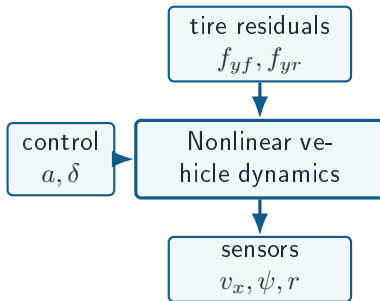
$$L_{\text{NN}} = P^{-1}Q^\top, \quad \lambda_{\text{NN}} = \sqrt{\sigma}.$$

Case study: nonlinear vehicle with unknown tire forces

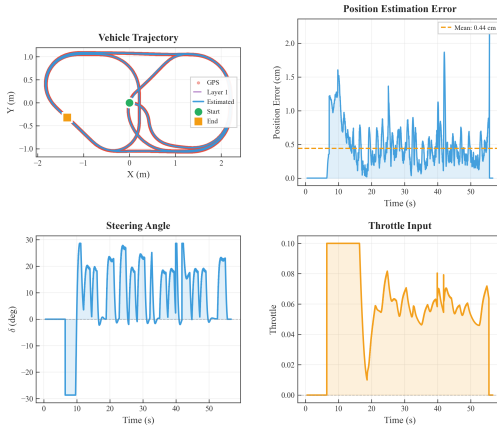
Vehicle state, input, and measurement

$$\begin{aligned} x &= [v_x \quad v_y \quad \psi \quad r]^T, & u &= [a \quad \delta]^T, \\ y &= [v_x \quad \psi \quad r]^T, & \mu_x &= [f_{yf} \quad f_{yr}]^T. \end{aligned}$$

- ▶ Lateral velocity v_y is not directly measured.
- ▶ Front/rear lateral tire-force residuals represent the unknown input.
- ▶ A layered controller provides the manoeuvring input used by both estimators.



Trajectory tracking: centimetre-scale position error



Observed outcome

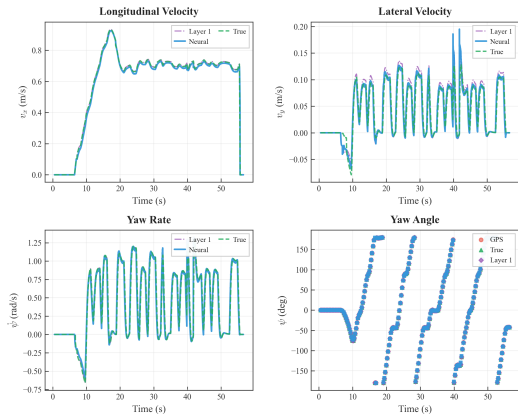
The estimated trajectory closely follows the GPS reference during the manoeuvre.

Reported metric

Mean position error: **0.44 cm**.

The lower panel confirms feasible steering and acceleration commands.

State estimation: the unmeasured lateral velocity is reconstructed

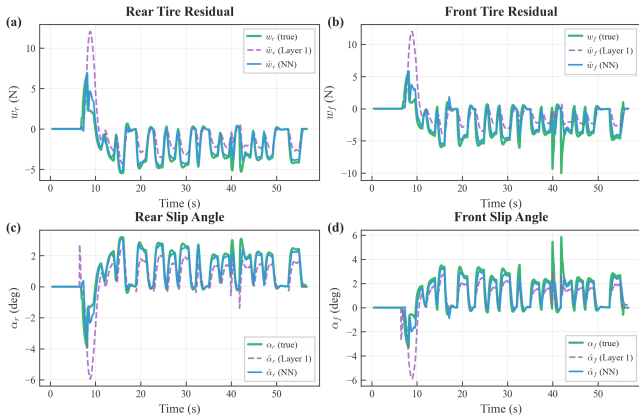


What matters

- ▶ Measured states remain consistent with their sensor channels.
- ▶ The estimator supplies v_y , which is absent from y .
- ▶ The learned observer refines transient performance beyond its certified teacher.

Residual learning improves tire-force reconstruction

Tire Force and Slip Angle Estimation



Interpretation

- Layer 1 gives a bounded, physically meaningful residual estimate.
- Layer 2 converges faster and tracks tire-force residuals more closely.
- The learned map captures state-dependent residual dynamics.

Takeaways

1. **Relaxed UIO design:** regularization and output derivatives remove reliance on exact unknown-input decoupling.
2. **Certified robustness:** LMs provide an explicit $\mathcal{H}^1$ bound for the generalized observer and an $\mathcal{L}_2$ bound for the neural observer.
3. **Reliable learning:** the UIO supplies stable supervisory signals; the neural observer learns the residual dynamics online.
4. **Vehicle evidence:** accurate trajectory tracking, recovery of the unmeasured lateral state, and improved tire-force estimates.

Main message

Certification and learning are complementary: the observer makes adaptation trustworthy, and adaptation makes the observer more accurate.